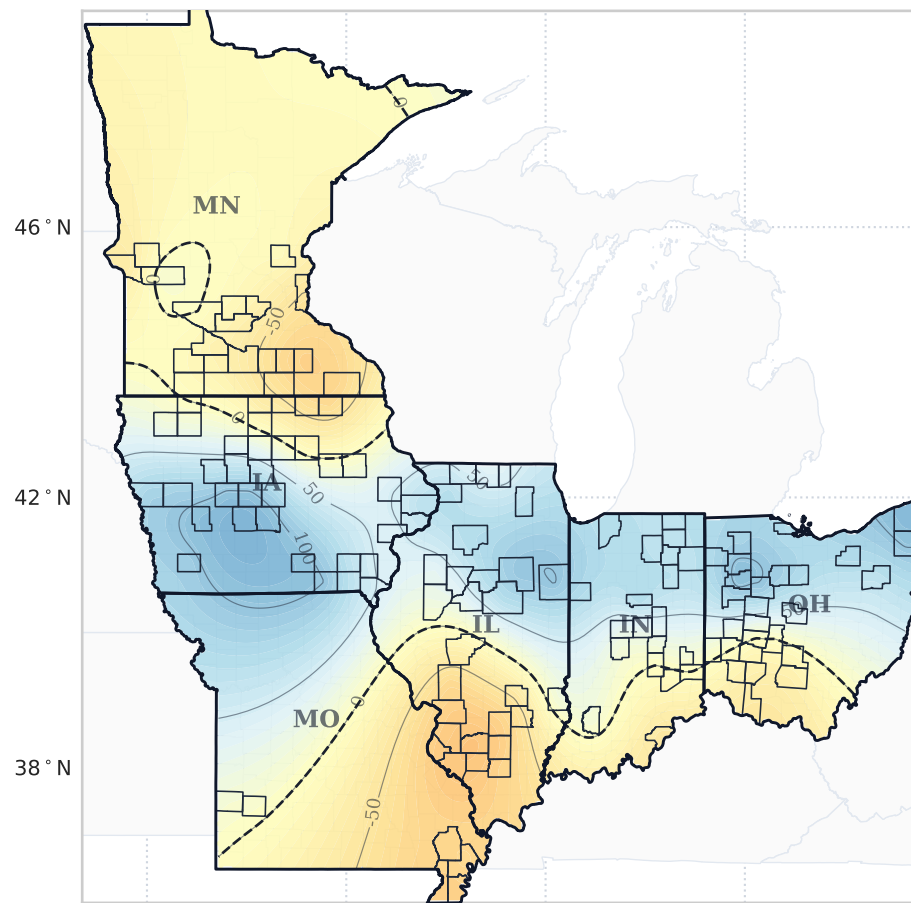
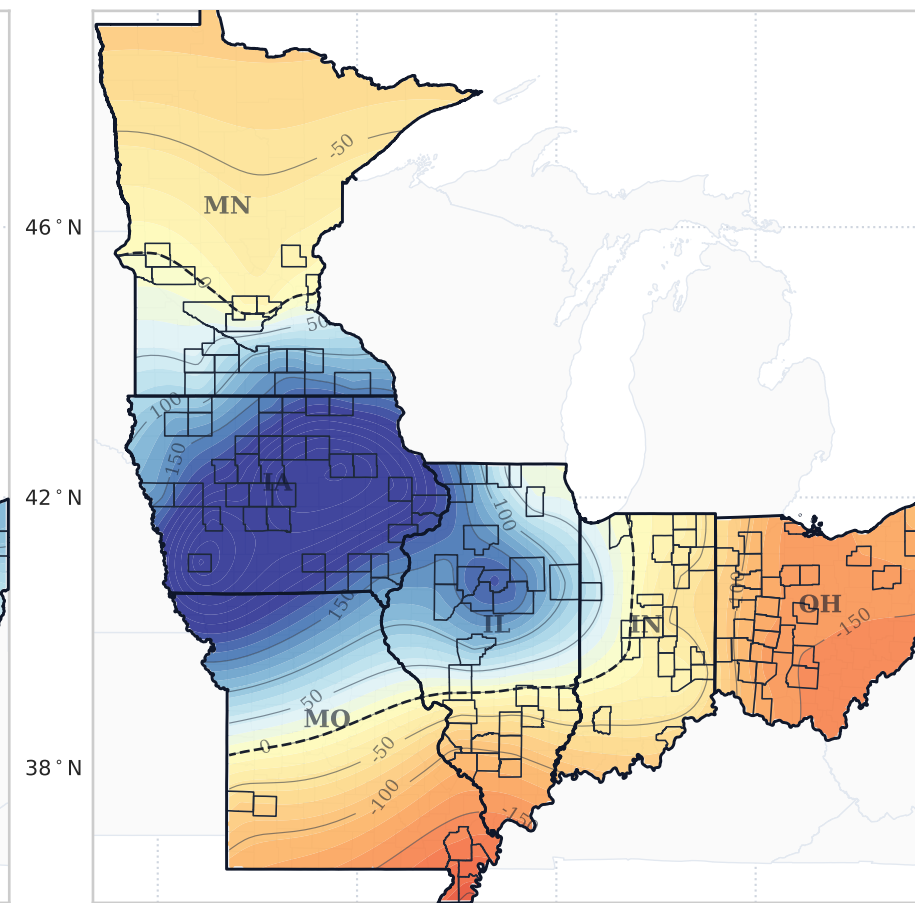


(a) 1992 Normal Reference Year (Yield: +2.1%)



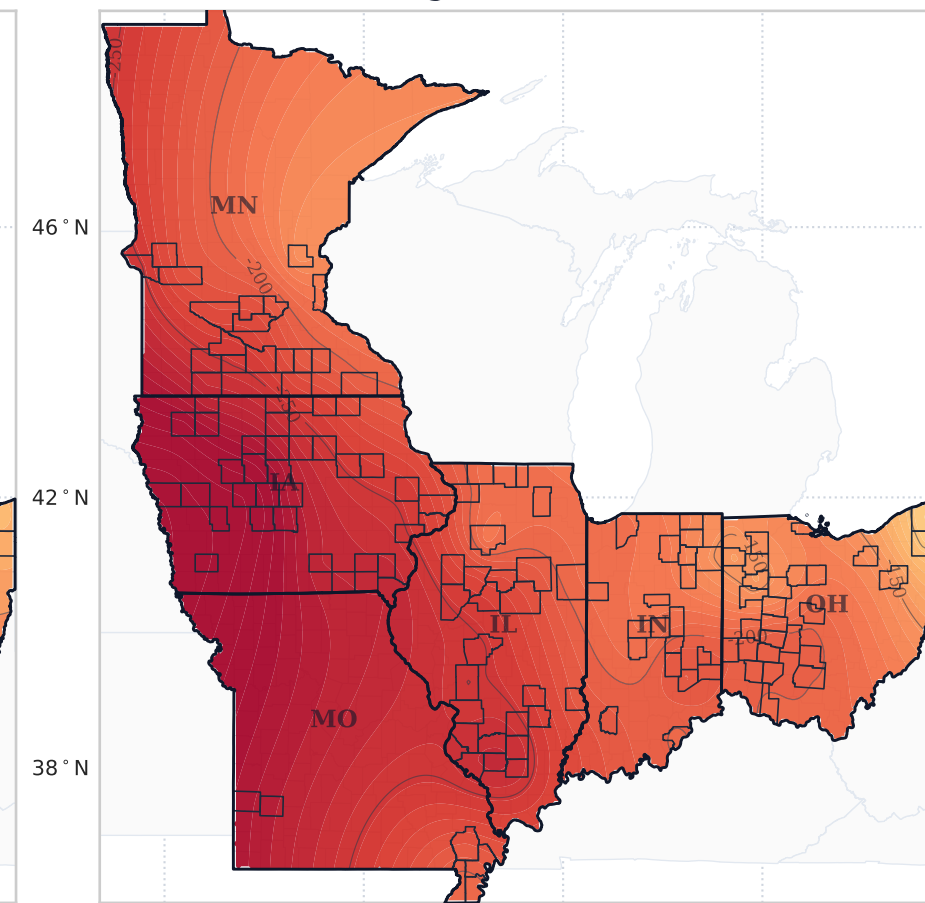
Near-average water balance across the study region

(b) 1993 Great Midwest Flood (Yield: -10.5%)



Persistent pluvial inundation & root-zone anoxia

(c) 2012 Flash Drought & Heatwave (Yield: -10.2%)

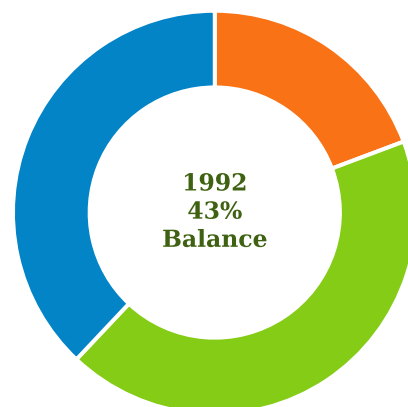


Acute evaporative surge & heatwave in Central/Southern belt

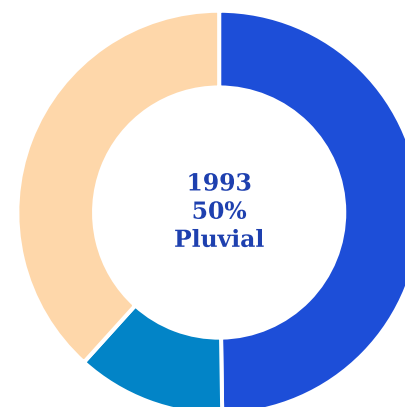


July--August Net Climatic Water Balance: $P - ET_0$ (mm) [Negative = Deficit / Drought $\leftarrow \rightarrow$ Positive = Surplus / Waterlogging]

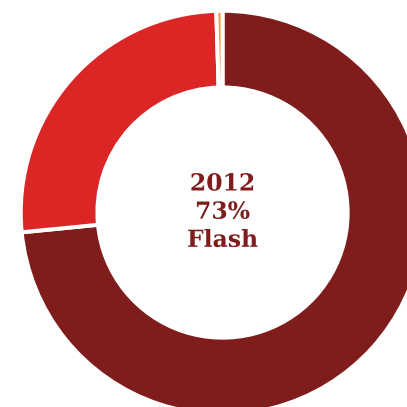
**1992 Acreage Exposure
(11.8M Harvested Acres)**



**1993 Acreage Exposure
(12.0M Harvested Acres)**



**2012 Acreage Exposure
(12.1M Harvested Acres)**



Harvest Exposure

Share of Midwest harvested soybean acreage exposed to climatic water balance:

- Severe Deficit (< -150 mm)
- Moderate Deficit (0 to -150)
- Excess Saturation (> +50 mm)
- Moderate Surplus (0 to +50)